

PC-04 · MOISTURE & FILM FORMERS · PERSONAL CARE

HS-D Panthenol

Panthenol

INCI	CAS	EC	FAMILY	SUPPLIED AS
Panthenol	81-13-0	201-327-3	PC-04	Viscous liquid, ≥98%

01 POSITIONING

One of the few humectants with decades of published clinical data behind the claim.

02 HOW IT WORKS

- 01

Penetrating humectant

Low molecular weight lets it cross the stratum corneum and bind water in depth, not just at the surface.
- 02

Provitamin conversion

Enzymatic conversion to pantothenic acid feeds barrier-repair metabolism.
- 03

Hair flexibilizer

Penetrates the cortex, raising flexibility and reducing breakage in mechanical stress tests (class data).

03 APPLICATIONS

- Moisturizers and after-sun
- Baby care (diaper area)
- Repair shampoos and conditioners
- Soothing and post-procedure lines
- Nail and lash care

04 FORMULATION GUIDE

USE LEVEL	0.5–5%
PH & CONDITIONS	Formulate pH 4–7; avoid prolonged heat above 70 °C

- Add in the cool-down phase; the D-form (dexpanthenol) is the biologically active one.
- Combines synergistically with glycerin and hyaluronate for layered hydration claims.

05 TYPICAL SPECIFICATION

Appearance	Clear viscous liquid
Assay	≥ 98%
Optical rotation (D-form)	+29° a +32°
Heavy metals	≤ 10 ppm

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Skin hydration uplift at 1–5% (class data)

D-panthenol systems



TEWL reduction

With panthenol



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.